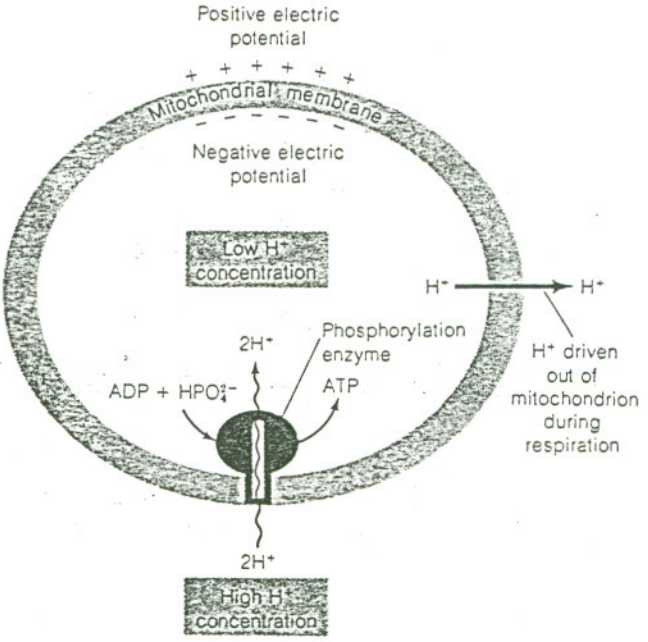
ANALYTICAL CHEMISTRY CUMULATIVE EXAM

November 16, 1991

Living cells convert energy derived from sunlight or combustion of food into energy-rich ATP (adenosine triphosphate) molecules. For ATP synthesis, $\Delta G = +34.5$ kJ/mole. This energy is then made available to the cell when ATP is hydrolyzed to ADP (adenosine diphosphate). In animal, ATP is synthesized when protons pass through a complex enzyme in the mitochondrial membrane.

Two factors account for the movement of protons through this enzyme into the mitochondrion (see figure below). One factor is the gradient arises because protons are pumped out of the mitochondrion by enzymes involved in the oxidation of food molecules.

A second factor is that the inside of the mitochondrion is negatively charged with respect to the outside. The synthesis of one ATP molecule requires two protons to pass through the phosphorylation enzyme.



Question 1: The difference is free energy when a molecule travels from a region of high activity to a region of low activity is given by:

$$\Delta G = -RT \ln \frac{a_{high}}{a_{low}}$$

$$R = 8.314 \text{ J/mol·K} \qquad T = 298 \text{ K}$$

How big must the pH difference be if the passage of two protons is to provide enough energy to synthesize one ATP molecule?

Question 2: pH differences this large have not been observed in mitochondria. How great an electrical potential difference between inside and outside is necessary for the movement of two protons to provide energy to synthesize ATP? In answering this question neglect any contribution from the pH difference.

Question 3: It is thought that the energy for ATP synthesis is provided by the pH difference and the electrical potential. If the pH difference is 1.00 pH unit, how many millivolts must be the potential difference?